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Narrative review

Optimization of antiviral dosing in Herpesviridae encephalitis: a promising approach to improve outcome?

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ABSTRACT

Background: Despite established antiviral therapy for herpes simplex virus, varicella zoster and cytomegalovirus encephalitis, the outcome remains poor.
Objectives: To assess pharmacokinetic (PK) and pharmacodynamic (PD) data of antiviral drugs in the central nervous system (CNS) to optimize the treatment of Herpesviridae encephalitis.
Sources: PUBMED search 1950 to September 2024, terms (1) "encephalitis" and ("HSV" or "VZV" or "CMV") or (2) cerebrospinal and ("(val)acyclovir" or "(val)ganciclovir" or "foscarnet" or "cidofovir").
Content: Antivirals against herpes viruses apparently act in a time-dependent manner. To suppress viral replication, drug concentration in the extracellular space at the site of the infection should be kept above the concentrations active in cell cultures for 24 h/d. Most data reflect delayed drug entry into lumbar cerebrospinal fluid (CSF). Ratios of the areas of the concentration/time curves (AUC) in CSF and serum (AUC_{CSF}/AUC_S) of acyclovir, ganciclovir and foscarnet are 0.15–0.3 in the absence of meningeal inflammation and increase in severe meningoencephalitis. Elimination half-lives ($t_{1/2\beta}$) are longer in CSF than in plasma. CSF concentrations are rough approximations of drug concentrations in the cerebral extracellular fluid. Lumbar CSF concentrations are usually higher than ventricular or cisternal CSF concentrations tending to overestimate cerebral extracellular fluid concentrations. Provided the availability of adequate measurements in individual CNS compartments, antiviral concentrations and efficacy may be predicted by future PK/PD modelling. Probenecid holds the potential to reduce the efflux of antiviral drugs from the CNS.
Implications: The long $t_{1/2\beta}$ of antiviral drugs in CSF suggest relatively uniform steady state CSF levels at dosing intervals $\leq$12h and accumulation after repeated dosing. Probenecid is of unproven utility. To rapidly attain effective concentrations in the infected tissue, physiologically based pharmacokinetic and PK/PD modelling may be helpful. Until reliable PK/PD data are available, doubling the first dose should be considered. **Roland Nau, Clin Microbiol Infect 2025;31:534**
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Introduction

Herpes simplex virus (HSV), varicella zoster virus (VZV) and cytomegalovirus (CMV) encephalitis occur worldwide and account for the majority of all viral encephalitis cases [1–4]. Intravenous acyclovir led to a reduction in the mortality of HSV encephalitis to approximately 20%. However, overall efficacy remains unsatisfactory with an unfavourable outcome (Glasgow outcome score, 1–4) in 70% of cases. Approximately 25% of patients develop autoimmune encephalitis with anti-neuronal antibodies [4,5]. Adjunctive immunomodulatory treatment has not improved outcomes [4]. Therefore, searching for more effective treatment strategies by optimizing dosing or introducing new antivirals continues.

The blood–brain barrier and blood–CSF barrier are indispensable for the adequate function of the CNS. However, they are also barriers for many anti-infective drugs to reach effective concentrations in the individual intracranial compartments (Fig. 1) [6–10].

Viruses replicate within host cells. Adequate concentrations of an antiviral drug must be present at all sites containing active viruses. The pharmacokinetic–pharmacodynamic (PK/PD) relationship is

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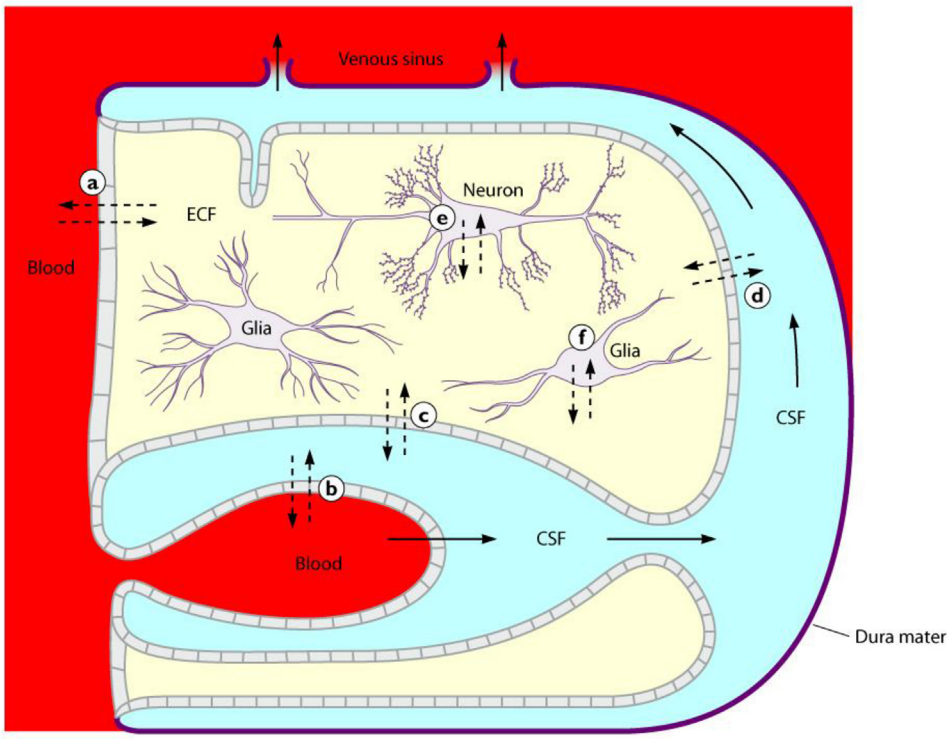


Fig. 1. Compartments in the central nervous system. The blood–brain and blood–CSF barrier can be considered a lipid bilayer surrounding the central nervous system (CNS) with leaky regions comprising approximately 1:5000 of the entire capillary surface area and with several outward and inward transport systems. In encephalitis, an antiviral drug has to pass several lipid layers before it reaches its target site, the intracellular space of the CNS [6–9]. The convoluted shape of the CSF space and the CSF flow impede the rapid equilibration of drugs among the different parts of the CSF compartment [6,7,9]. In CSF, generally, small molecules equilibrate more readily than large ones [9]. Lumbar CSF concentrations tend to overestimate cerebral ECF levels after intravenous or oral drug administration [7,9]. The direction of the CSF flow is illustrated by continuous arrows. Interrupted arrows indicate the directions of the diffusion of water or solutes between brain capillaries, CSF and nervous tissue: across (a) the blood–brain barrier, (b) the epithelium of the choroid plexus, (c) the ventricular ependyma, (d) the pia-glial membranes at the surface of the brain and spinal cord, (e) and (f) the cell membranes of neurons and glial cells [6]. The border between ventricular CSF and the extracellular space of the brain (c) (the "inner CSF–brain barrier," anatomically represented by ependymal cells joined by gap junctions) in adults allows even large molecules to pass and does not constitute a major obstacle to the entry of drugs into the extracellular space of the brain [6]. Drug concentrations in ventricular and cisternal CSF can be considered rough approximations of drug levels in the adjacent ECF. The concentrations of blood-derived compounds usually increase from the ventricles to the lumbar CSF space [6,9]. The border between the subarachnoid space and the extracellular space of the brain (d) (the "outer CSF–brain barrier") restricts diffusion by a radial glial end feet layer covered with a surface layer of the pia mater. The thick line represents the dura mater and arachnoidea surrounding the whole CNS (reproduced from [6,7] with the kind permission of Elsevier and the American Society for Microbiology). Cells also are surrounded by lipid layers with active transport pumps. Compared to the transporters at the barriers, drug transporter proteins at the cellular level have been studied less thoroughly. *In vitro*, the intracellular concentrations of antiviral drugs in astrocytes, microglia, pericytes and brain microvascular endothelial cells differed strongly [9,10]. CSF, cerebrospinal fluid.

further complicated by the uncertainty of whether antiviral concentrations effective *in vivo* can be deduced from those active *in vitro*.

Methods

This review will discuss the PK/PD relationships of drugs well-established for the treatment of viral encephalitis and will address the question of whether changes in dosing regimens may improve outcomes. We performed a PUBMED search from 1950 to September 2024 with the search terms (1) "encephalitis" and ("HSV" or "VZV" or "CMV") or (2) (cerebrospinal or toxicity) and ("(val)acyclovir" or "(val)ganciclovir" or "foscarnet" or "cidofovir"). Antiretroviral drugs are not covered in this review.

Pharmacodynamics of antiviral drugs

Traditionally, antiviral activity has been studied in animal models. Encephalitis models for HSV are well-established [11].

Since viruses need eukaryotic cells to replicate, often the activity of antiviral drugs is studied in cultures of eukaryotic cells as monolayers. The concentrations of the antiviral drugs studied to determine *in vitro* activity usually represent extracellular concentrations. Three-dimensional cultures mimicking human tissues containing the major neural cell types may reflect more realistically the *in vivo* conditions [12,13].

The determination of the viral load in CSF by quantitative PCR appears to be useful in the assessment of the response to treatment in CMV encephalitis [3] and in HSV-1 encephalitis [4,14,15].

Antiviral activity usually is characterized by the EC_{50}, which is the concentration at which inhibition of viral growth is 50% of the maximum response within the specified exposure time [16]. In organotypic raft cultures using human keratinocytes, the concentration of acyclovir necessary to reduce the viral cytopathic effect (HSV-1 and HSV-2) or the formation of plaques (VZV) by 50% in sensitive strains was approximately 0.2mg/L in HSV-1 and HSV-2 and 0.1mg/L in VZV [17]. At present, it is unclear, which multiple of the EC_{50} ensures maximum antiviral activity *in vivo*.

The lack of a clear PK/PD relationship is a major challenge in optimizing antiviral dosing for CNS infections. Limited data suggest that acyclovir and other nucleosides/nucleotide analogues act as time-dependent compounds. The PK/PD parameter, most suitable for time-dependent antivirals, appears the time in the extracellular fluid (ECF) at the site of infection above the EC_{50} [16,18]. For acyclovir, a time above the EC_{50} of $\geq$50% of the dosing interval at the site of infection is considered the minimum for an effective therapy [19–21]. Models for the dose–concentration–response relationship for optimal treatment of VZV and CMV infections were

developed from data generated in infected immunocompromised patients establishing differential equations for the rate of change of acyclovir/ganciclovir concentrations in individual compartments and for the changes of VZV/CMV viral load in plasma [22,23]. These PK/PD models yielded relatively high mean concentrations inducing half-maximum killing: acyclovir 10.5mg/L *in vivo* vs. 0.8–4.0 mg/L *in vitro*; ganciclovir 13.9mg/L *in vivo* vs.0.01–9.5 mg/L *in vitro*. Compared to plasma trough concentrations, ganciclovir AUC_{0-24h} better predicted the decline of CMV load in plasma, and conventional dosing of (val)ganciclovir appeared inadequate to reduce the CMV load rapidly [23]. With the aid of more data on the concentrations and antiviral activity of acyclovir and ganciclovir in CSF and brain ECF, these models could be expanded to the treatment of HSV, VZV and CMV encephalitis.

Although acyclovir and valacyclovir in general are well tolerated, dose-dependent adverse effects (particularly renal failure, leukopenia and neuropsychiatric symptoms) preclude large increases of the daily dose [20,24]. Ganciclovir and foscarnet frequently cause myelosuppression and renal failure, particularly, in patients with a pre-treatment impaired renal function [18].

Pharmacokinetics

Systematic studies on the pharmacokinetics of drugs established for the treatment of herpesvirus encephalitis in CSF are scarce (Table 1) [25–37]. When we failed to identify appropriate data in humans, we reported studies in non-human primates [38].

Acyclovir and valacyclovir

Central compartment

Acyclovir is a deoxynucleotide analogue of guanosine with a plasma protein binding of 9–33% independent of the plasma concentration. In virus-infected cells, it is rapidly converted by kinases to the active acyclovir triphosphate. Despite dose adjustment, trough acyclovir concentrations and trough and peak concentrations of the main metabolite 9-carboxymethoxymethyl guanine were higher in patients with abnormal than normal renal function [20]. Carboxymethoxymethyl guanine has no antiviral activity but has been associated with neuropsychiatric symptoms in patients treated with (val)acyclovir [24,39].

The prodrug valacyclovir contains a valine group that strongly increases oral bioavailability. Following oral uptake, it is nearly completely hydrolyzed to acyclovir and L-valine by first-pass metabolism in the gut and liver [39].

Central nervous system

The pharmacokinetics of acyclovir in CSF has been incompletely studied (Table 1). After 1 month of oral acyclovir treatment, two patients suffering from multiple sclerosis received a lumbar catheter, and blood and CSF were drawn repeatedly after dosing. Despite fluctuating serum concentrations, CSF concentrations remained stable at approximately 0.11 mg/L. A re-estimation from the original Figure 3 [25] yielded an approximate area under the concentration-time curve in CSF/ area under the concentration-time curve in serum (AUC_{CSF}/AUC_S) of 0.18 and 0.4, respectively. $T_{1/2\beta CSF}$ was long. In three multiple sclerosis patients receiving repeated lumbar punctures, serum acyclovir ranged from 0.21 to 1.42 mg/L (mean, 0.56 mg/L) 1.5 hours before and from 0.53 to 1.41 mg/L (mean, 0.92 mg/L) 1.5 hours after dosing. Again, the mean CSF concentrations were stable at 1.5 hours before dosing and 1.5 hours after dosing (difference not statistically significant) suggesting a long $t_{1/2\beta}$ in CSF [25]. In a neonate with HSV-2 meningoencephalitis, CSF penetration was excellent [26] (Table 1).

Because of its favourable pharmacokinetics, valacyclovir has been suggested as an acceptable early oral treatment for suspected HSV encephalitis in resource-limited settings [40]. In six subjects with normal renal function and three with impaired renal function (doses: Table 1) with indwelling lumbar CSF catheters, CSF and plasma were collected repeatedly after the fifth dose. CSF C_{max} concentrations were close to the average CSF concentrations and reached 2.3 and 5.3 hours after dosing, and mean AUC_{CSF}/AUC_{plasma} were 0.270 in normal and 0.276 in reduced renal function (Fig. 2A) [27]. Similarly, in 10 multiple sclerosis patients with normal renal function and CSF-to-serum albumin ratio receiving 1 g valacyclovir every 8 hours, the estimated AUC_{CSF}/AUC_S based on measurements in lumbar CSF 2 hours and 8 hours after dosing was 0.19 ± 0.03 [28]. Nevertheless, intravenous acyclovir must be preferred for the following reasons: (1) After oral valacyclovir, in HIV-infected clinically stable volunteers, plasma concentrations in mean peaked after 2.0–2.5 hours [41], whereas after i.v. acyclovir, plasma concentrations peak at the end of the infusion (i.e. at 1 hour). Since the time to reach effective CNS concentrations is a main determinant of outcome, any delay must be avoided. (2) In critically ill patients, gastrointestinal absorption often is impaired. Therefore, accelerated intravenous to oral switch programmes are discouraged [42].

In a rat model without inflammation, in the prefrontal cortex, the mean $AUC_{unbound,\ brain\ ECF}/AUC_{unbound,\ blood}$ of acyclovir studied by microdialysis was 0.316 and 0.427 in the absence and presence of the organo-anion transporter-3 (OAT3) inhibitor probenecid, respectively. The $AUC_{unbound,\ brain\ ECF}$ more than doubled by probenecid treatment because OAT3 inhibition both reduced acyclovir renal clearance and removal from the CNS by efflux pumps [43].

Treatment of HSV/VZV encephalitis with increased (val)acyclovir doses or prolonged treatment protocols

The administration of acyclovir 15 mg/kg instead of 10 mg/kg every 8 hours did not improve outcomes in adult patients [1] but was associated with more frequent renal failures [44,45]. Treatment of adults with oral valacyclovir 2 g 3x daily for 3 months after a standard course of intravenous acyclovir did not improve outcomes [46].

In neonates with general and CNS HSV infection, high-dose acyclovir (3 × 20 mg/kg/d) was more effective than the standard or medium doses (30 or 45 mg/kg/d) [26,47]. Reversible neutropenia occurred in 6 of 29 newborns receiving 60 mg/kg/d. No other severe adverse effects were noted [47].

Ganciclovir and valganciclovir

Central compartment

Ganciclovir is a small hydrophilic compound with a low oral bioavailability and low plasma protein binding. It is mainly excreted renally. In healthy adults, $t_{1/2\beta}$ in plasma is approximately 3.5 hours, with a strong increase in renal failure [48]. In critically ill patients, the variation of pharmacokinetic parameters is high. The prodrug valganciclovir, a valine ester, is well absorbed from the gastrointestinal tract and rapidly metabolized to ganciclovir (with 65% oral bioavailability). Co-administration with the OAT3 inhibitor probenecid increased the $AUC_{plasma\ 0-8h}$ of oral ganciclovir by approximately 50% [49].

Central nervous system

In adult rhesus monkeys, ganciclovir i.v. produced an AUC_{CSF}/AUC_{plasma} ratio of 0.155 ± 0.071 (Fig. 2B) [38].

Table 1
Antiviral drugs established in the therapy of human herpesviruses (values of adults with normal renal function and approximately 70 kg body weight, unless indicated otherwise)

Drug molecular weight; XLogP3-AA	Number, type of patients, CSF sampling site	Single doses							Repeated doses or continuous infusion (CI)									Reference
		Dose (mg)[a]	C_{CSF} other than C_{max} (mg/L) hours after the end of infusion	C_{tissue} (mg/L)	CL (L/h)	V_d (L)	$t_{1/2\ CSF}$ (h)	$t_{1/2S}$ (h)	Dose (mg/d)[a]	$C_{max\ CSF}$ (mg/L)	C_{ssCSF} Meningeal inflammation (mg/L)	C_{ssCSF} Uninflamed meninges (mg/L)	C_{CSF} other than C_{max} (mg/L) hours after dosing	$C_{ss\ plasma}$ (mg/L)	Cl (L/h)	V_d (L)	AUC_{CSF}/AUC_S or C_{ssCSF}/C_{ssS}	
Aciclovir 225.2 g/mol; −1.9	9–20, A, L, uif	2.5–15/kg			15.6[b]	48[b]		3.16 ± 0.20	3 × 2.5–15/kg				3.7, 4.0; 2 h	6.7–20.6[b]				[36]
Aciclovir 225.2 g/mol; −1.9	7, A, L, uif								3 × 800 po			0.16 (0.11–0.24) 0.21 (0.14–0.28)					0.31 (0.13–0.52); approx. 0.18, 0.40[c]	[25]
Aciclovir 225.2 g/mol; −1.9	1, N, L, ifl								3 × 20/ kg				4.8, 6.2; 4.5 h, 2 h				0.67, 0.71[d]	[26]
Valaciclovir (aciclovir level measured)	10, A, L, uif								3 × 1000 po			0.56 ± 0.20 0.52 ± 0.16			35 ± 6[e]	110 ± 20	0.19 ± 0.03	[28]
Valaciclovir (aciclovir level measured)	6, A, L, uif, normal renal function								3 × 2000 po	1.76 ± 0.34		1.41			45.5 ± 6.0[e]		0.27	[27]
Valaciclovir (aciclovir level measured)	3, A, L, uif, chronic kidney disease								2 × 1500 po	3.60 ± 1.10		3.00			8.7 ± 3.2[e]		0.28	[27]
Ganciclovir 255.2 g/mol; −2.5	3, A & C, L, ifl				17.5 ± 9.1[b]	81 ± 38[b]		3.60 ± 1.40[b]	2 × 5/kg				0.70 (0.56–0.79), 0.08 (0.03–0.17), 0.45 (0.27–0.72)	5.3 ± 2.8[b]				[30]
Valganciclovir (ganciclovir measured)	1, N, L, ifl								2 × 16/kg				0.52					[31]
Valganciclovir (ganciclovir measured)	1, A, ifl	900 po		2.60[f]	38.9[e]		4.5[f]	3.2									0.69[f]	[29]
Foscarnet 126.0 g/mol; −2.0	5, A, L, uif								202–274/kg			8.63 ± 1.57			11.7 ± 4.52	42.0 ± 23.3	0.43 ± 0.19	[33]
Foscarnet 126.0 g/mol; −2.0	21 & 5, A, L ifl	90/ kg	131 ± 52 1 h						180/kg		308 ± 155[g]						0.23 ± 0.16[h] 0.66 ± 0.11[i]	[34]
Foscarnet 126.0 g/mol; −2.0	27, A, L, uif	56–213/ kg	10.1 (0–63.0) 0–12 h														0.27 (0–3.4) 0–12 h/d	[32]
Foscarnet 126.0 g/mol; −2.0	3, A, L, ifl								40–100/kg			7.59 (2.21–15.37)						[35]

A, adult; AUC, area under the concentration-time curve; CSF, cerebrospinal fluid; ifl, inflamed; L, lumbar CSF, N, neonate; po, per mouth; uif, uninflamed, V, ventricular CSF; XLOGP3-AA = method to estimate the octanol–water partition coefficient (log P) by adding up contributions from each atom in the given molecule [37]; molecular weights and XLOGP3-AA values are taken from PUBCHEM (https://pubchem.ncbi.nlm.nih.gov/).

[a] Intravenous administration, if not indicated otherwise.
[b] Data derived from reviews concerning the pharmacokinetics of antiviral drugs [21,36,48].
[c] AUC estimated from the original Fig. 3(a) and (b), n = 2 [25].
[d] Neonate with highly permeable neonatal blood–CSF barrier.
[e] Apparent oral clearance (CL/F).
[f] Extracellular fluid of a glioblastoma patient after surgery.
[g] Concentrations higher than in comparable studies.
[h] Single dose.
[i] Repeated dosing.

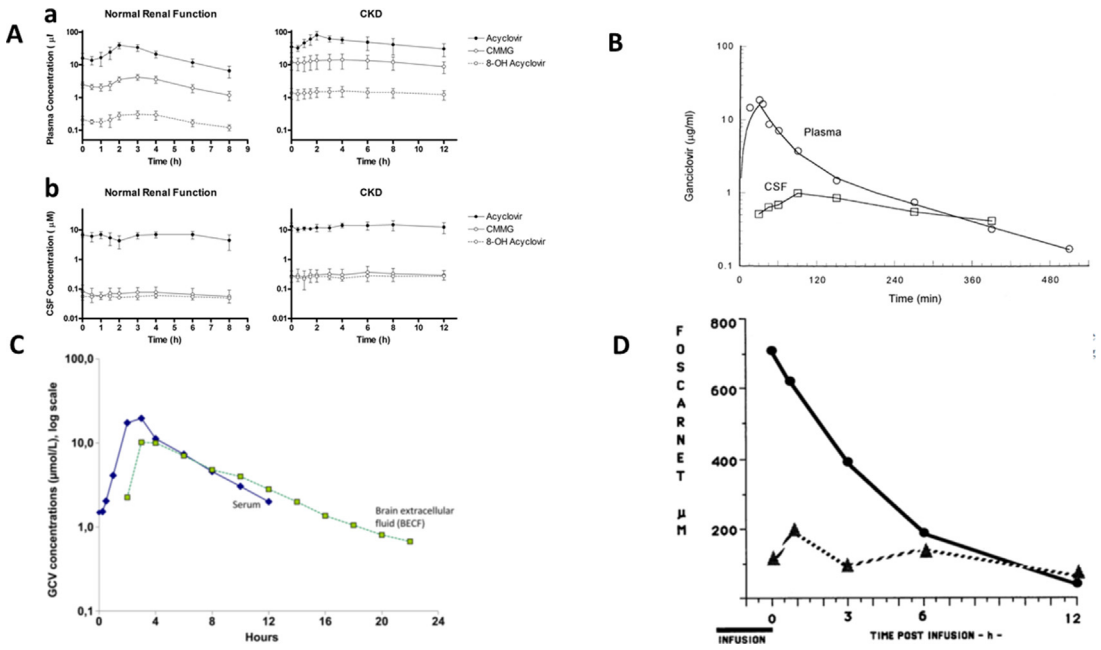


Fig. 2. Penetration of antiviral drugs into the CSF and brain extracellular fluid of humans or non-human primates. (A) Plasma (a) and lumbar CSF (b) concentration–time profiles of acyclovir and its metabolites in six adults with normal (studied after the fifth dose of 2000 mg valacyclovir orally every 6 hours) and three participants with impaired renal function (studied after the fifth dose of 1500 mg valacyclovir orally every 12 hours) (means ± standard deviations). Please note the very stable CSF concentrations precluding the estimation of the elimination half-life ($t_{1/2\beta}$) in CSF (reproduced from [27] with the kind permission of the American Society for Microbiology). (B) Concentration–time curves of ganciclovir in plasma (circles) and ventricular CSF (squares) in an adult male rhesus monkey following an i.v. dose of 10 mg/kg ganciclovir administered over 30 min. The mean area under the concentration-time curve (AUC) of ganciclovir in ventricular CSF was 2.80 ± 1.38 mg/L x h, and the mean peak CSF concentration was 0.7 ± 0.3 mg/L. The elimination half-life ($t_{1/2\beta}$) in plasma was 1.8 hours and in CSF (estimate from the figure) approximately 6 hours. Please note the crossing of the concentration–time curves typical for hydrophilic drugs in the absence of strong meningeal inflammation. The long $t_{1/2\beta}$ in CSF ensures relatively uniform CSF and cerebral extracellular fluid concentrations during treatment (reproduced from [38] with the kind permission of Springer Nature). (C) Ganciclovir concentrations in the dialysate (open squares) and in serum (filled rhombs) in a 59-year-old man treated repeatedly with valganciclovir (900 mg/d) after resection of his glioblastoma and placement of a microdialysis catheter in the adjacent tissue of the resected tumour. Please note that the concentration–time curves in serum and cerebral extracellular fluid almost run in parallel as a consequence of strong impairment of the blood–brain barrier in the vicinity of the resected glioblastoma (reproduced from [29] with kind permission of BMJ Publishing Group). (D) Mean foscarnet concentrations in plasma (circles) and lumbar CSF (triangles) determined at various times after the end of an infusion of foscarnet [56–213 mg/kg of body weight (median 100 mg/kg) over 2–6 hours] (median, 3 hours) in HIV-infected adult patients. Please note the relatively uniform CSF compared to the fluctuating plasma concentrations and the slow elimination of foscarnet from CSF (reproduced from [32] with the kind permission of the American Society for Microbiology). CSF, cerebrospinal fluid.

In a rat glioma model using microdialysis, the AUC of unbound ganciclovir in blood ECF, brain ECF and tumour ECF were 26.19, 7.05 and 20.56 mg/L x h yielding average AUC ratios of the unbound drug of 0.269 in normal brain and 0.785 in tumour ECF [50]. Similarly, in a patient with a microdialysis catheter adjacent to the resected glioblastoma, oral valganciclovir 900 mg/d resulted in maximum serum and ECF concentrations of 5.0 and 2.6 mg/L, an elimination half-life in serum and ECF of 3.2 and 4.5 hours, and an AUC_{ECF}/AUC_S of 0.69 (Fig. 2C, Table 1) [29].

In three adult patients with HHV-6 encephalitis or myelitis treated with ganciclovir, the median CSF concentrations drawn 1 hour after dosing were 0.70, 0.08 and 0.45 mg/L. In one patient, therapy was continued with oral valganciclovir 2 × 900 mg/d; the concentration of ganciclovir in CSF remained largely unchanged [30].

In an infant with CMV infection treated by oral valganciclovir, CSF concentrations at steady state were approximately 1/10 of the peak plasma and close to the trough plasma concentrations [31] (Table 1).

Foscarnet

Foscarnet is a hydrophilic low-molecular-weight inhibitor of herpesvirus DNA polymerase for i.v. administration [4].

As a consequence of its small molecular size, foscarnet readily enters the CSF with CSF-to-plasma concentration ratios >0.2 in most cases. In AIDS patients, after an infusion of over 2–6 hours, the median penetration based on the determination of concentrations in individual samples was 0.27. CSF concentrations were higher in the presence of meningeal inflammation, and the CSF-to-plasma concentration ratio increased with the time of sampling after the end of the infusion [32] (Fig. 2D). During a continuous infusion, CSF drawn on day 2–13 contained therapeutic foscarnet concentrations, and the CSF-to-serum concentration ratio was 0.43 ± 0.19 ($n = 6$) [33]. One hour after a single dose of 90 mg/kg foscarnet over 2 hours, the CSF-to-serum ratio was 0.23 ± 0.16 ($n = 21$). After 4 weeks of therapy, the CSF-to-serum ratio was 0.66 ± 0.11 ($n = 5$) [34]. In three patients with HHV-6 encephalitis, therapeutic foscarnet concentration in CSF during treatment with 40–90 mg/kg every 12 hours was achieved [35] (Table 1).

Other antiviral drugs established in the treatment of viral encephalitis

The entry of cidofovir, an acyclic nucleoside analogue (molecular weight 279.2 g/mol, XLogP3-AA -3.6), into human CSF has not been studied. Because of its hydrophilicity and slightly larger molecular size, it most likely penetrates less readily than acyclovir, ganciclovir and foscarnet. Although cidofovir shows *in vitro* activity against HSV, VZV, CMV, Epstein–Barr virus, and HHV-6, it is not used for the treatment of encephalitis by human herpesviruses. In contrast, brincidofovir, an orally bioavailable lipid conjugate of cidofovir, readily cross the blood–brain and blood–CSF barrier, and alone and in combination with acyclovir is active in animal models of encephalitis [4].

Discussion

Under the assumption that nucleoside/nucleotide analogues act in a time-dependent manner, concentrations in all infected compartments ideally are 24 h/d above the EC_{50} in cell cultures. This suggests frequent dosing of antiviral drugs in patients with normal renal function [45]. Since their protein binding is low, alterations in protein binding have a low impact on the entry of acyclovir, ganciclovir and foscarnet into the CNS. The concentration–time curves in CSF and probably also in the other CNS compartments lag behind those in serum/plasma, i.e. effective concentrations in CNS are not attained immediately [7,27,32,38] (Fig. 2). The small hydrophilic compounds used to treat Herpesviridae encephalitis in the absence of strong meningeal inflammation possess a CSF penetration (assessed as CSF–serum ratios of steady state concentrations or AUC_{CSF}/AUC_{plasma}) of approximately 15–30%. The slow elimination of hydrophilic antiviral drugs suggests moderate to strong drug accumulation in the CNS. Since the rapid achievement of effective antiviral drugs appears to be crucial for a favourable outcome, a higher loading dose, e.g. doubling of the first intravenous dose, should be evaluated clinically. Co-administration of the OAT3 inhibitor probenecid increases the AUC of acyclovir and ganciclovir in plasma and inhibits their elimination from the CNS rising CNS concentrations by both mechanisms [43,49]. Inhibition of the efflux pumps P-gp, Bcrp, Mrp2, and Oat3 increased the entry of acyclovir into the CNS [43]. Whether efflux pump inhibition may improve outcomes remains to be studied.

The best *in vivo* correlates of the drug concentrations tested in cell-culture-based assays, at present, are the drug concentrations in the ECF of infected tissue. ECF drug concentrations can be measured in the brains of animals and—when intensive monitoring including intracranial pressure measurement and surveillance by microdialysis is clinically indicated—also in patients with encephalitis. When microdialysis is not indicated, the assessment of CSF trough concentrations of the active compound, toxic metabolites and the viral load in CSF appears promising for pharmacokinetic surveillance [3,15].

PK/PD modelling may optimize dosing in encephalitis [8]. For the establishment of valid models, at present (1) not enough human pharmacokinetic data (in particular, ventricular/cisternal CSF and brain ECF measurements) to construct valid concentration–time curves in these compartments by a population kinetic approach and (2) not enough pharmacodynamic data linking CSF and brain ECF concentrations to the rate of viral killing are available. Microdialysis in experimental animals with simultaneous measurement of drug concentrations in several compartments of the CNS, and—provided there is a clinical indication—in patients, is the key to PBPK modelling of drugs in the ECF of human CNS [51–54].

Since for ethical reasons continuous PCR determination of the viral load in cerebral ECF and in neural cells in patients is almost impossible, positron emission tomography with [^{18}F]fluoro-3-hydroxy-2-propoxy)methyl]guanine ([^{18}F]FHPG) as virus tracer may be able to study pharmacodynamics in human HSV encephalitis [55]. More tracers for molecular imaging of the viral load by positron emission tomography (PET) or single photon emission computer tomography (SPECT) will become available to guide optimization of treatment [56].

Neurofilament light chain (NfL) is an established biomarker of axonal injury. In HSV encephalitis, CSF NfL levels predicted long-term neurocognitive outcomes [57], and in VZV encephalitis, serum NfL helped to estimate neuronal injury [57,58]. MicroRNAs encoded by HSV and by the injured brain measured in CSF may monitor disease progression and therapy [59]. Other biomarkers detectable in blood of potential value in viral encephalitis are S100B, glial fibrillary acidic protein (glial cell damage) and neuron-specific enolase [60]. It remains open whether repeated measurements of plasma concentrations of these biomarkers can monitor the efficacy of antiviral therapy.

Conclusions

The *in vivo* concentration–effect relationship of antiviral drugs essentially is unclear. The concept that antiviral drugs act in a time- and not in a concentration-dependent manner should be confirmed by adequate cell culture and animal experiments.

Lumbar CSF concentrations of systemically administered drugs are usually higher than ventricular or cisternal ones and tend to overestimate cerebral ECF concentrations. Brain ECF concentrations are difficult to assess in patients. Antiviral drugs may not reach sufficiently high values for treating viral encephalitis in all relevant compartments. PK/PD modelling will open new avenues for dose optimization when sufficient information on the PD and the ECF concentrations of antivirals are available. Until then, doubling the first dose may be an option to rapidly attain effective concentrations in the infected tissue.

Transparency declaration

Potential conflict of interest

The authors declare that they have no conflicts of interest.

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